C PROGRAMMING FROM SCRATCH · EPISODE 02

Choosing Your C Tools

Compilers, Linkers and IDEs

KEY POINTS

- The processor runs binary machine instructions; C source is text, so C is a compiled language.
- Compile turns each .c file into an object file; link joins them with libraries into an .exe.
- A compiler (MSVC, GCC, Clang) is one tool; an IDE is the workspace around it.
- Our setup is Visual Studio 2026 Community with the Desktop development with C++ workload.

01 Text In, Binary Out

▶ VIDEO 00:36

- The processor runs **machine instructions** (binary).
- C **source code** is plain text for people, so it must be translated first.

> KEY IDEA **C is a compiled language**
>
> Before a C program runs, its source code is translated into machine instructions.

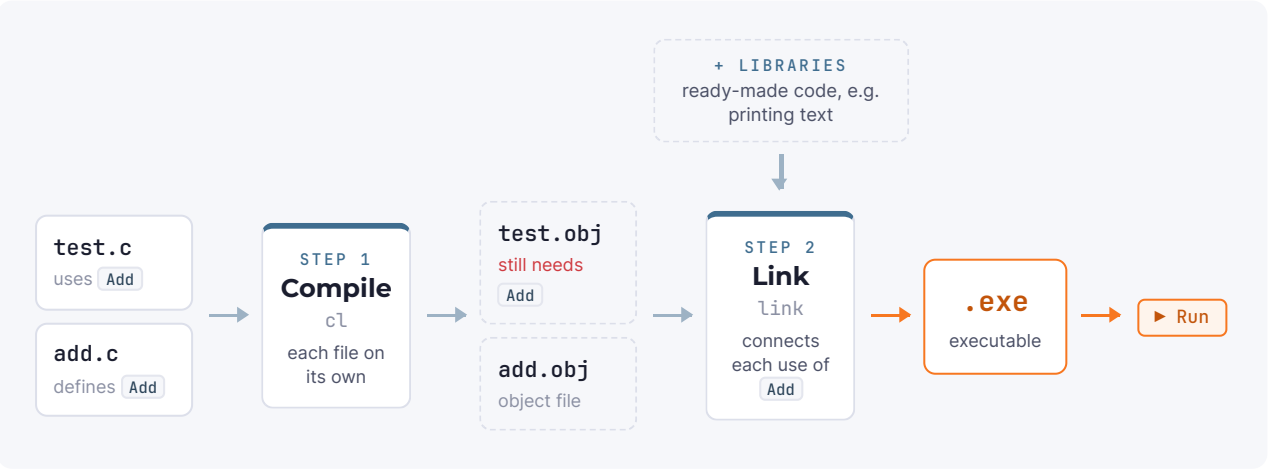
02 Compile, Then Link

▶ VIDEO 01:32

A program is often split across several `.c` files. Two steps turn them into one program:

- **Compile**: the compiler turns each source file, on its own, into an **object file**.
- **Link**: the linker joins the object files and **libraries** into one **executable**.

FIGURE 2.1 **From source files to a program that runs**



`test.obj` holds machine instructions but isn't a finished program: it still needs `Add`. The linker joins the object files and libraries into the one file that actually runs.

The two steps also tell you where an error comes from:

ERROR FROM THE…	IS USUALLY ABOUT…
compiler	the code in a file
linker	pieces it can't find

03 Compilers and IDEs

▶ VIDEO 03:36

- **Compiler toolsets**: **MSVC** (Microsoft; compiler `cl`, linker `link`), **GCC** (GNU, common on Linux), **Clang** (LLVM; used by Apple's tools).
- An **IDE** puts the editor, the build tools and the debugger in one application. Clicking **Build** runs compile and link for you.

04 A Tour of the Options

▶ VIDEO 05:41

IDE	RUNS ON	COMPILER	WORTH KNOWING
Visual Studio	Windows	MSVC	Large install, but simple setup; little extra configuration
Xcode	Mac	Clang	Apple's IDE; the natural choice on a Mac
Code::Blocks	Cross-platform	GCC	Free and lighter; some downloads need the compiler set up yourself
Dev-C++	Windows	GCC	Small, quick to install; basic features, last release 2021
CLion	—	Your choice	JetBrains; works with CMake; free for non-commercial use

> WATCH OUT **Visual Studio ≠ Visual Studio Code**
>
> **Visual Studio** is a full IDE. **Visual Studio Code** is an editor: C support comes from extensions, and the compiler is installed separately.

05 Our Setup

▶ VIDEO 07:35

- **Visual Studio 2026 Community**: free for individuals and classrooms; editor, MSVC and debugger in one place. Drawback: it needs much more disk space.
- In the installer, tick only the **Desktop development with C++** workload. It covers C too.
- Visual Studio 2022 also works; on other setups the C is the same.